

2005 - 2016 SAPS CRIME ANALYSIS

PRE-PLAN DOCUMENT

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Role: Data Analyst (Portfolio Project)

PROJECT OVERVIEW

This project analyses South African Police Service (SAPS) crime statistics from 2005 to 2016 to understand crime patterns across provinces and identify crime categories experiencing significant growth. The objective is to move beyond raw crime totals and evaluate crime burden relative to population size while identifying long-term crime trends that may require greater policy attention.

Key objectives

To analyze provincial crime statistics and crime category trends in order to identify areas of elevated crime burden and crime categories showing sustained growth over time.

Problems identified

- Raw crime totals can be misleading when comparing provinces with different population sizes.
- High-population provinces naturally record more crimes, making risk assessment difficult.
- Crime trends across multiple years are difficult to identify from raw tables.
- Growth in specific crime categories may remain hidden without structured analysis.
- Decision-makers lack summarized visual insights to support interpretation.

Target Audience

- **Primary audience:** Public safety stakeholders, Community safety departments and Policy analysts.
- **Secondary audience:** Hiring managers and Data analysts.
- **Tertiary audience:** General public and Researchers.

User story

A public safety stakeholder wants to understand which provinces experience the highest crime burden and which crime categories are growing most rapidly so that crime patterns can be interpreted more accurately, and long-term risks can be identified.

Use cases

1. PROVINCIAL CRIME BURDEN ANALYSIS

Analyze crime rates across provinces using population-adjusted metrics.

Acceptance Criteria

The analysis should:

- Calculate crime rates per 100,000 residents.
- Compare all nine provinces.
- Rank provinces by crime burden.
- Identify highest-risk and lowest-risk provinces.
- Present results visually.

2. CRIME TREND ANALYSIS

Analyze crime statistics over multiple years.

Acceptance Criteria

The dashboard should:

- Compare annual crime figures across years.
- Identify increasing and decreasing trends.
- Highlight significant shifts over time.
- Present trends using charts and tables.

3. CRIME CATEGORY GROWTH ANALYSIS

Determine which crime categories experienced the strongest growth between 2005 and 2016.

Acceptance Criteria

The dashboard should:

- Calculate percentage growth by crime category.
- Rank categories by growth rate.
- Highlight categories experiencing exceptional increases.
- Present results visually.

4. PROVINCIAL COMPARISON ANALYSIS

Compare crime exposure between provinces.

Acceptance Criteria

The analysis should:

- Enable province-to-province comparisons.
- Highlight differences in crime burden.
- Identify provinces above and below the national average.
- Present findings clearly through charts.

Success criteria from user perspective

The solution is successful if it can:

- Easily Identify provinces facing the highest crime burden.
- Compare provinces fairly using population-adjusted metrics.
- Highlight long-term crime trends.
- Identify rapidly growing crime categories.
- Support evidence-based interpretation of crime statistics.
- Present findings in a clear and understandable format.

Information needed from user perspective

Example:

- Total crimes by province
- Population by province
- Crime trends over time
- Growth of crime categories
- Provincial crime comparisons

Data needed from analyst perspective

The raw fields required to support include:

Example schema:

Field	Data Type
Province	Text
Crime Category	Text
Crime Count	Integer
Year	Integer
Population	Integer

Data quality checks

Measures to ensure the dataset is complete, accurate, and usable for analysis:

- Row count validation
- Missing value checks
- Duplicate record checks
- Province validation
- Crime category validation
- Population data validation
- Year continuity validation
- Percentage calculation verification

Additional requirements

- Document analysis methodology.
- Document population-adjustment calculations.
- Present findings visually.
- Maintain reproducible workflows.
- Store project files and documentation in GitHub.
- Provide conclusions and recommendations



